Multi-Head Attention

Deep Learning

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Contents of This Video

In this video, we will cover:

- Limitations of single-head attention
- Multi-head attention architecture
- How heads learn different patterns
- Concatenation and final projection
- Typical configurations

The Single-Head Limitation

One attention head produces one attention pattern

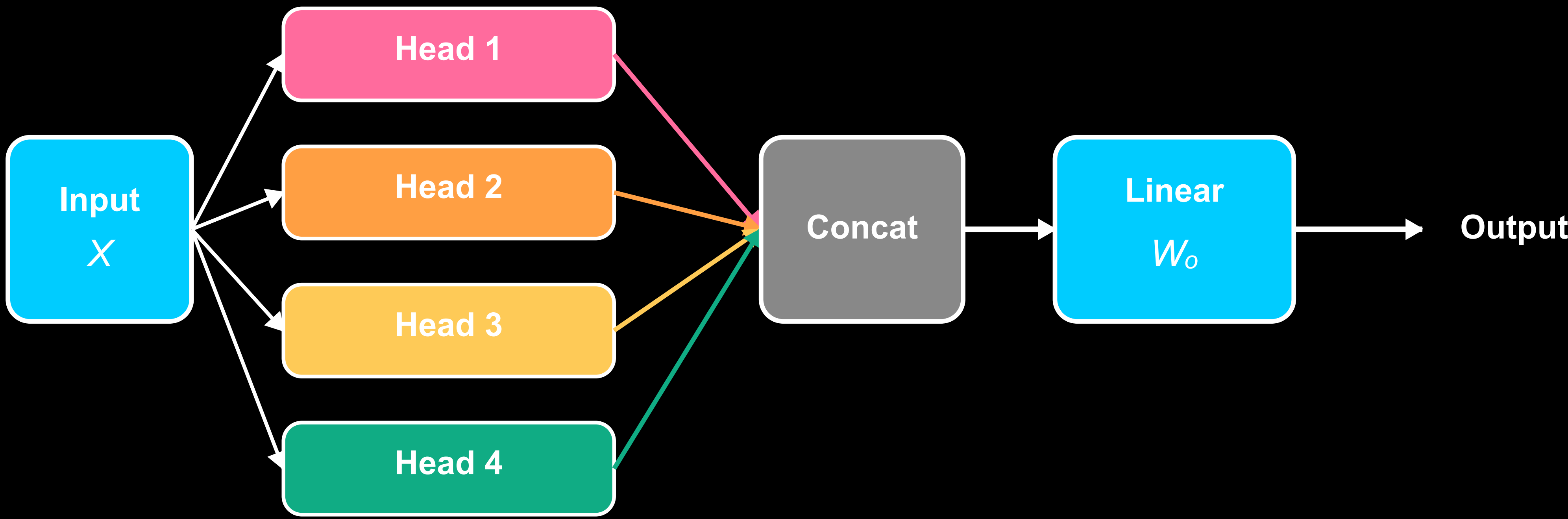
But words have multiple relationships:

- “The cat sat” → syntactic (subject-verb)
- “cat … it” → coreference
- “sat … mat” → semantic (location)

Single head must choose what to focus on

Multi-Head Attention

Multi-Head Attention: Multiple Parallel Attention Operations



Multi-Head Formula

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W_O$$

Where each head:

$$\text{head}_i = \text{Attention}(QW_Q^i, KW_K^i, VW_V^i)$$

Parameters:

- h heads, each with $d_k = d_{model}/h$ dimensions
- Total parameters similar to single large-head attention

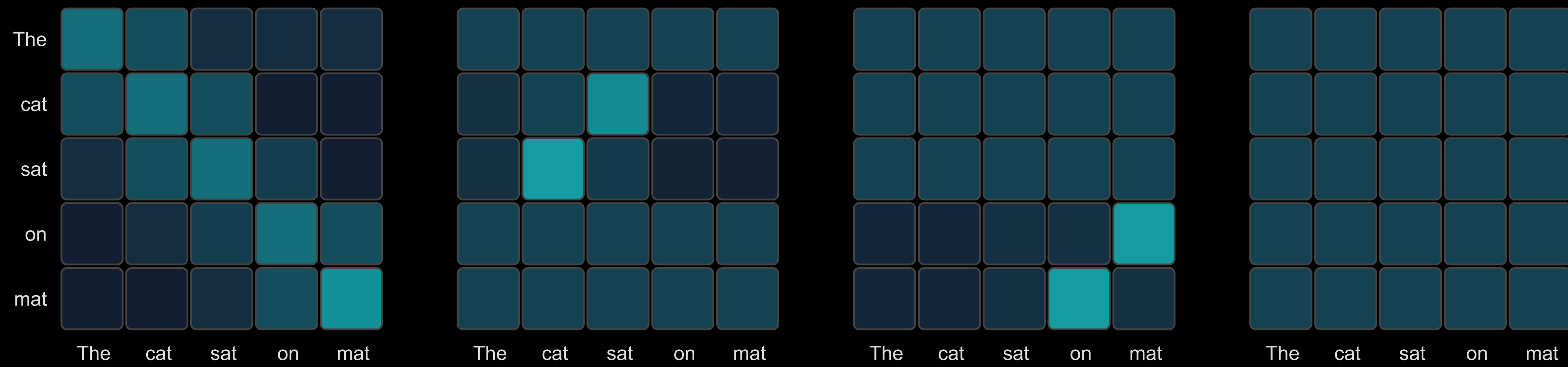
What Different Heads Learn

Head 1: Adjacent words

Head 2: Subject-verb

Head 3: Prep-object

Head 4: Global avg



Typical Configurations

Model	d_{model}	Heads (h)	d_k per head
BERT-base	768	12	64
BERT-large	1024	16	64
GPT-2	768	12	64
GPT-3	12288	96	128

Common pattern: $d_k = d_{model}/h$ = 64 or 128

What We've Covered

In this video, we've learned:

- Single attention head captures only one relationship type
- Multi-head: run multiple attention operations in parallel
- Each head has its own Q, K, V projections
- Heads often specialize in different relationships
- Concatenate outputs and apply final linear transformation
- Typical: 8-16 heads with 64 dimensions each